these characters; the generator suggests sample texts when one is missing. Works for PDF (read w
p the key like a password: with key and work, anyone can read the message; without the key,
data in a work A spreadsheet or data file can travel inside a work and be read back out of the
data_model with a checksum. `"sh node cli/tg-data.mjs encode people.xlsx -o people.t
brary, command line, tests and the web are done. In the browser, the generator's panel "Dat
and opens a read dates with a key. Everything happens in the browser, nothing is up
cs/encrypted-data.md](https://typographics-art.org/docs/encrypted-data.md)). ### Sealed
es an argument. See [docs/encryption-data.md](https://typographics-art.org/docs/encrypted
vault (the secret alone recovers everything; a strict 2-of-3 mode exists), with Argon2id for the e
hy a different cipher is not the answer). A plain-language page for
d) (German: [docs/passwords-and-keys.de.md](https://typographics-art.org/docs/password
today's cryptography is known", not "nobody can ever get at it": a lost key loss
ted media and the adapter.** A picture or a video can be sealed so it only opens with the ke
without the plaintext ever touching the disk and without anything being uploaded. Large me
me, size). See [docs/encrypted-media-adapter.md](https://typographics-art.org/docs/
w-internal-2026-10.md); a second review by a different model, done blind to the first, is in [docs/
ple layout positions in, so no rendered hash round them.
S ** 3 units/page at its default resolution
`scripts/legibility.mjs` + `scripts/legibility.py`
nails reads the title in some, not all, cases, so
ibility: mod and results Typographics Art el
A **human** sees the *image layer*: the artw
directly — not OCR'd from the pixels — so it is exac
he hard case, and it is why the image layer i
`, recomputes it from the glyphs and requires it to eq
* tg verify` fails the moment the data layer and the im
ast never hide instructions from the people who see th
* (per letter) and *word* (per word); Roboto Flex, phot
t equality with the source | | PDF, stored order | `pdftote
between source and OCR output | Python `difflib`, divided
act | exact | 97.3 % word order | 92.7 % | 98.0 % | | free • cha
ic, Opus 5.5), reading images directly. - **Material:** one cro
op] that reading was discarded. - **Procedure:** ** transcriptio
ops to 32 % at *free*, the language-aware model read everythin
output (`GLYPH_CSS`, in `src/render.js`). That also keeps the layo
er the full text layer=recognition still moves some
characters counted, exact text equality checked for MuPDF, whic
4 %) | Poppler `pdftotext -r 600` | 10 of 10 apart from line-en
r example " –" in Roboto Flex) fro
that was wrong. A plain page of
, social media), use legibility *high
Not measured yet - More visio
ize, measured # File size: what m
193 | 0.939 / 0.923 | 0.25 % | | witho
e width | 1.92 MB | 40 | 0.949 / 0.935 | 1.24 % | | 1 size, one width | 1.65 MB | 20 | 0.963 / 0.9
, one width: only colour or grey is left | 0.85 MB | 2 | 0.911 / 0.900 | 4.03 % | ### Basic Law
ccessibility tags | 0.61 MB | 22 | 0.995 / 0.995 | 0.00 % | | one width (no condensed/expand
31 % | | 1 size, one width | 0.72 MB | 13 | 0.894 / 0.894 | 0.43 % | | colour step 40, one width |
t | 0.60 MB | 2 | 0.961 / 0.961 | 5.12 % | What this shows: - **One type size is sharper.** Mixi
contrast. With neither weights nor sizes left, only colour draws the picture and likeness dro
p in a PDF. One-pagers therefore use **one size and no width variation** by default (`cli/resp
s # prints every setting, writes docs/file-size-study.json (about 45 minutes) node scripts/size
ly its public text, nothing invisible, no extra characters. The message lives only in the key. ## H
secret.js`). The same text comes out of the HTML, the SVG and the PDF (read with MuPDF). 2
A-256 of the public text. Revealing with another work, or with a changed text, is detected and re
of up to two million characters). The key grows by about 4 characters per character of the secret
y node cli/tg-secret.mjs reveal work.pdf --key tg1.... # prints the message ``` The key is the secret,
(via MuPDF), SVG, HTML and plain tex
re is no hidden layer to find, because
peat. - **Not a replacement for encryp
s digits and rare symbols may be abs
text then differs from its source and
nd-keys.md --> ## Passwords, keys
vered. Everything is open source;
cs/key-management.md) and [cipher-review.
does not match.
web generator has the same under
y does and does not protect: [docs/hidden-mes
an `.xlsx`, `.csv`, `.tsv`, `.json`, `.md`, `.yaml`, `.xml`
node cli/tg-data.mjs decode work.pdf --format csv # rea
ata`) attaches a file, shows what will be written, and either writ
e files are in `samples/data/`. Still command line only; passwords
so be **sealed**: AES-256-GCM, with a key (`tgx1....`, 64 characters) tha
[docs/key-format.md](https://typographics-art.org/docs/key-format.md). K
d faceter. **A password never is** the only secret of a public record.** AES-256-G
rts explains the password rule, the three recommended setup
s.de.md)). **Is this secure?** The cipher (AES-256-GCM, random 256-bit key) is sou
data, a compromised device defeats everything, and a password alone neve
an **adapter** (`cli/tg-adapter.mjs`) makes the sealed file readable by *any* ordina
a seekable `tgst1` stream file (the one-pager is its printable cover); a small image or a d
ypted-media-adapter.md). An intern
urity-review-internal-2026-10-b.md](htt
a real text: HTML, SVG, or PDF with a t
yph-On in every weight) P(Photo) --> D(T
widths). 2. **Map.** The photo becomes
e font size at which the whole text fills th
pick some of 28 sample texts) and drop a p
ssets/fonts/fonts.css`, so opening such
s from the work's HTML, SVG and PDF). #
s 100–1000) legibility: 'medium', // high |
// HTML snippet, text as characters toSv
and line Needs Node 22+ (the CLI uses th
cli/tg-art.mjs photo.jpg --text poem.txt --
u send, paid with a DigiByte transaction (
Hokusai | <de>Friedrich</de> Nietzsche • C. D. Friedrich | <b
gallery](https://typographics-art.org/ga
pographics-art/typogram/blob/main/NOT
self.** A poem made of the sister's blue
DA, JAWS or VoiceOver. - **Findable art.**
range text along a shape, usually by han
ts?** A weight axis (and sometimes a w
you use. ## Contributing Typographics
om/typographics-art/typogram/blob/ma
n run dev # needs Python 3, `fonttools`, ga
| the library | | `cli/` command-line too
e by `scripts/build-samples.mjs`) | | `
The Typographics Art community. co
Art** (short **TG Art**) is the technique o
Rules A work is Typographics Art when it
ay leave areas blank and widen the gap at a space (or indent a line) so that glyphs stand
s reduced instead. A longer token (a URL, a base64 string, the dot-joined lines of a data
t that keeps the characters in text: HTML, SVG, or a PDF with text layer. - Every glyph
and so nearly one per glyph; concatenated they equal the text, and no word is split ac
TML `role="figure"` label). Exported HTML and SVG keep the text once, in the set for
), - typeface, - colour, - weight and - optical size. - **Machine-
ne works (tens of thousands of characters) are read by zooming in or by mach
rs exactly its declared text and nothing invisible; the message lives only in
ference Implementation works The pipeline lives in `src/`. Simply by
ll three rules. ### 1. Fidelity - Every character
project's own texts and models an
ure from a given text by setting that t
- **coverage**: the share of the glyph's cell (advance × line he
would silently measure a fallback font. ## 2. Tone map (`
= (t − target)² • 0.00035·(1 − size) • 0.0
For every character, a table maps (t
outAt` Linear interpolation between
ams words like "office", which matters for ligat
, whitespace normalised). The test su
only | `tesseract` on a PNG rendered
—|—|—| High | 97.3 % | exact | exact |
ge model (first sample) Classic OCR k
the word order cannot be known in a
um | 68 | 100 % | 94 | 100 % | Th
s in PDFs.** Chrome set "fil" in *offic
text sizes, and `pdftotext`'s layout m
ones read a one-pager compl
ding order | | pdfminer.six | 10 of 10 |
ny chars limit"?): one-pagers above that lose most of their tex
ne type size (no glyphs below the base size), this holds even
le characters are identical; the build reports such a page as
SVG.** The text is exact at every setting. PDF is exact too wi
With Poppler, pass a high resolution (`pdftotext -r 600` or mo
ens of thousands of characters, where glyphs are a few pixe
ont sizes, and `pdftotext`'s layout m
ones read a one-pager compl
ding order | | pdfminer.six | 10 of 10 |
| all weights, widths and sizes, colo
weights, one width | 0.58 MB | 12 | 0.9
weights, 1 size, one width | 0.54 MB | 4
tead of 0.940) at a third of the size of t
g measurable and saves 12 %. - **Coa
and shows the expected PDF size. We
--> ## Hidden messages # Hidden mes
a machine reads the work: whitespace normalised,
and can be chosen at random (Web Crypto, without
d, are the secret. ## The key `tg1.` followed by ba
of 106 characters: a 30-character secret gives abo
history: `reveal` also takes it from a file (`--key-fil
`revealSecret(publicText, key)` from `src/index.js
dom positions:** the same secret gives a differen
ython. ## Index * A character that does not occ
t makes it invalid. - One PDF quirk: after a chara
hnique.md): the work contains its declared text
protects data, without reading the cryptogra
ments behind it are [encrypted-data.m
- **The lock is strong.** Sealed data u